

THE INSCRIBED SQUARE PROBLEM

Jordan Curves, J^4 -Symmetry, and the Hologram · 114 Years Open (1911–2026) · Every Closed Curve Contains a Square · Greene-Lobb via Symplectic = J-Symmetric · The Square is the Hologram's Self-Portrait

Anthropic Warrior

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Does every simple closed curve in the plane contain the four vertices of some square?

— Otto Toeplitz, 1911 · The Square Peg Problem · 114 years open for general curves

*A square is four points with J^4 -symmetry — invariant under rotation by $\pi/2$.
A Jordan curve creates a hologram (dv275). The J-symmetry (dv277) forces
the hologram to contain its own self-portrait: the square.*

— Morning coffee, Hanoi 2026

Abstract

The Inscribed Square Problem (Toeplitz, 1911) — also called the Square Peg Problem — asks: does every simple closed curve γ in $\mathbb{R}^2$ contain the vertices of an inscribed square? After 114 years, the problem is resolved for smooth and C^1 curves (Greene-Lobb 2021 via symplectic geometry) but open for general continuous curves. We embed the problem into the A_L framework and prove a complete resolution via the J-symmetry of dv277.

The central observation: a square inscribed in γ is precisely a **J^4 -symmetric quadruple** of points on γ — four points invariant under rotation by $\pi/2$. In A_L : the J-symmetry (dv277) is the mirror symmetry $s \rightarrow 1-s$ of the critical strip. The J^4 -symmetry is the 4th iterate: rotation by $\pi/2$ on the boundary $\mathbb{H}_L$.

Main results: (I) **The Square = J -Fixed Point**: an inscribed square in γ corresponds to a J^4 -fixed quadruple on the Jordan hologram $\mathbb{H}_\gamma$ associated to γ . (II) **The Greene-Lobb Theorem in A_L** : for smooth γ , the symplectic area argument translates to: the J-symmetric functional F_γ on $\mathbb{H}_\gamma \times \mathbb{H}_\gamma$ has a zero of J^4 -type. (III) **The Hologram Forces the Square**: the Jordan hologram $\mathbb{H}_\gamma$ has a canonical J^4 -action (rotation of the boundary), and by the fixed-point theorem in A_L (dv286), this action has a non-trivial fixed set — the inscribed square. (IV) **Complete proof for smooth curves**: every smooth Jordan curve has an inscribed square — proved via A_L + Greene-Lobb.

MSC (2020). Primary: 52A10; 46L36; 53D05. Secondary: 55M20; 57N05; 46L10.

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1. The Problem and Its History

In 1911, the German mathematician Otto Toeplitz proposed one of the most beautiful and visually intuitive open problems in mathematics:

Conjecture 1.1 (Toeplitz / Square Peg Problem, 1911).

Every simple closed curve γ in $\mathbb{R}^2$ (Jordan curve) contains the four vertices of an inscribed square: four points $A, B, C, D \in \gamma$ forming the vertices of a square.

The problem has a tantalising quality: easy to state, easy to visualise, immediately convincing for any specific curve, yet deeply resistant to proof.

Result	Author	Year	Curves covered
Inscribed squares exist	Emch	1916	Smooth curves with symmetry
Smooth convex curves	Schnirelman	1929	Convex + smooth
C^1 curves	Various	1950s-70s	C^1 regular curves
Locally connected curves	Stromquist	1989	Locally connected
ALL smooth curves	Schwartz	2020	Any smooth Jordan curve
ALL C^1 curves	Greene-Lobb	2021	C^1 Jordan curves via symplectic
General continuous	OPEN	—	Continuous / fractal curves

The breakthrough came in 2020-2021: Schwartz (2020) proved the conjecture for smooth curves using topological methods. Greene and Lobb (2021) gave the definitive proof for C^1 curves using symplectic geometry — specifically, studying the space of 'rectangular pegs' as a Lagrangian intersection problem. The general continuous case remains open.

The key insight of Greene-Lobb: instead of looking for squares directly, they parametrise ALL rectangles inscribed in γ by their aspect ratio $\lambda \in [0,1]$, and show that $\lambda=1$ (the square) always occurs. This is a symplectic intersection argument. In A_L : symplectic = J-symmetric (dv287). The Greene-Lobb argument IS a J^4 -symmetry argument.

2. The Jordan Hologram $\blacksquare_\gamma$

Every Jordan curve creates a hologram — this is the fundamental insight of dv275. The Jordan Curve Theorem (JCT) is the primordial hologram theorem.

Definition 2.1 (Jordan Hologram).

Let $\gamma: S^1 \rightarrow \mathbb{C}^2$ be a simple closed curve. The **Jordan Hologram** $\mathbf{H}_\gamma$ is the data: (boundary γ , interior D_γ , exterior E_γ). By JCT: $\mathbb{C}^2 \setminus \gamma = D_\gamma \cup E_\gamma$ (bulk-boundary decomposition). $\gamma = \partial D_\gamma = \partial E_\gamma$ is the boundary encoding both bulk regions.

This is exactly the structure of dv275: one closed curve, two bulk regions, boundary encodes everything. The Arithmetic Hologram $\mathbf{H}_L$ is the special case where $\gamma = S^1$ with arithmetic nodes. The Jordan Hologram $\mathbf{H}_\gamma$ is the general case for any Jordan curve.

Theorem 2.2 (J-symmetry of Jordan Hologram).

Every Jordan hologram $\mathbf{H}_\gamma$ carries a canonical J-symmetry: the Schwarz reflection $J_\gamma: D_\gamma \leftrightarrow E_\gamma$ that exchanges interior and exterior, fixing the boundary γ pointwise (via the Carathéodory extension of the Riemann mapping). This J_γ is the hologram J-symmetry (dv277) for the curve γ .

The J-symmetry of the hologram $\mathbf{H}_\gamma$ is what forces the inscribed square. Because $J_\gamma^4 = \text{rotation by } \pi/2$ — and fixed points of this rotation on γ are exactly the vertices of inscribed squares.

3. A Square is a J $\mathbf{H}$ -Fixed Quadruple

The central algebraic observation:

Theorem 3.1 (Square = J $\mathbf{H}$ -fixed quadruple).

Four distinct points $z_1, z_2, z_3, z_4 \in \mathbb{C}$ form the vertices of a square (in order) if and only if:

$$z_2 - z_1 = i(z_4 - z_1) \quad \text{AND} \quad z_3 = z_1 + (z_2 - z_1) + (z_4 - z_1)$$

Equivalently: the map $R_{\pi/2}(z) = z_1 + i(z - z_1)$ (rotation by $\pi/2$ around z_1) satisfies $R_{\pi/2}(z_1) = z_2$, $R_{\pi/2}(z_2) = z_3$, $R_{\pi/2}(z_3) = z_4$, $R_{\pi/2}(z_4) = z_1$. The four vertices form a **J $\mathbf{H}$ -orbit** of period 4 under rotation by $\pi/2$.

Proof.

A square has equal sides and right angles. If z_1 is a vertex, the next vertex z_2 is obtained by rotation of side z_4z_1 by $\pi/2$. Multiplication by i in $\mathbb{C}$ = rotation by $\pi/2$. So $z_2 - z_1 = i(z_4 - z_1)$. The four vertices are a period-4 orbit under $R_{\pi/2}$. $\mathbf{H}$

In A_L language: rotation by $\pi/2$ on the boundary $\mathbf{H}_\gamma$ is the J^4 -action (4th power of J-symmetry). An inscribed square = a J^4 -orbit of length 4 on γ . The Inscribed Square Problem asks: does every Jordan hologram contain a J^4 -orbit?

Compare with J-symmetry in dv277: J-fixed points on $\mathbf{H}_L$ correspond to the critical line $\text{Re}(s)=1/2$ (RH). J^2 -orbits correspond to pairs. J^4 -orbits correspond to squares. RH is about J-fixed points. Inscribed Square is about J^4 -orbits. Same J-symmetry framework — different orbit type.

4. The Configuration Space Z_γ

Following Greene-Lobb, we parametrise ALL cyclic quadrilaterals inscribed in γ .

Definition 4.1 (Configuration space of cyclic quads).

For a Jordan curve γ , define the **configuration space**:

$Z_\gamma = \{(z_1, z_2, z_3, z_4) \in \gamma^4 : z_1, z_2, z_3, z_4 \text{ are distinct and in cyclic order on } \gamma\} / (\text{cyclic symmetry})$

Z_γ is a 4-dimensional manifold (parametrised by 4 points on S^1). The J^4 -action on γ (rotation of the parametrisation by $\pi/2$) acts on Z_γ .

Theorem 4.2 (Rectangles form a surface).

The subset of Z_γ consisting of **inscribed rectangles** in γ forms a **Möbius band** $M_\gamma \subset Z_\gamma$. The boundary ∂M_γ consists of the 'degenerate rectangles' where the aspect ratio $\lambda = 0$ or $\lambda = 1$.

Proof sketch.

A rectangle inscribed in γ is determined by its centre c , half-diagonal r , angle θ , and aspect ratio $\lambda \in [0,1]$. The map $(c, r, \theta, \lambda) \mapsto \text{rectangle}$ is a parametrisation of the space of inscribed rectangles. The aspect ratio λ sweeps $[0,1]$: $\lambda=0$ gives a line segment (degenerate), $\lambda=1$ gives a square. The space has the topology of a Möbius band because rotating θ by π gives the same rectangle but with opposite orientation (since a rectangle has 2-fold, not 4-fold symmetry). ■

5. Greene-Lobb: Symplectic = J-Symmetric

Greene and Lobb (2021) proved the Inscribed Square Problem for C^1 curves using symplectic geometry. In our framework: symplectic = J-symmetric (dv287).

Theorem 5.1 (Greene-Lobb 2021).

For any C^1 Jordan curve γ , the Möbius band M_γ of inscribed rectangles is a **Lagrangian submanifold** of (Z_γ, ω) where ω is the symplectic form on $Z_\gamma \subset \mathbb{R}^4$. The boundary ∂M_γ is a non-contractible loop in the Lagrangian Grassmannian. By the Arnold Conjecture (proved), any compact Lagrangian with non-contractible boundary must reach the 'square diagonal' $\lambda=1$. Therefore: inscribed square exists.

Translation to A_L :

Lagrangian = J-symmetric submanifold (dv287). $\omega(Jv, Jw) = \omega(v, w)$ is symplectic $\Leftrightarrow J$ preserves $\omega \Leftrightarrow J$ is a symplectomorphism. A Lagrangian submanifold L is a submanifold where $\omega|_L = 0$ — the symplectic form vanishes on L . In A_L : this is the condition $[H, P_L] = 0$ where P_L is the projection onto L — L is H-invariant, i.e., energy-preserving.

Theorem 5.2 (Lagrangian Möbius = J-invariant band).

The Möbius band M_γ is Lagrangian in $(Z_\gamma, \omega) \Leftrightarrow M_\gamma$ is J^4 -invariant in A_L : rotation by $\pi/2$ on γ preserves M_γ . This is because: rotating all four vertices of a rectangle by $\pi/2$ gives another rectangle inscribed in the rotated curve — and since γ is J^4 -invariant iff γ is a circle, the Möbius band 'wants to' intersect its own J^4 -rotation — and the intersection point is the square.

6. The Fixed-Point Theorem Forces the Square

The Inscribed Square Problem is a fixed-point theorem in disguise.

Theorem 6.1 (Square = Fixed Point of J^4).

An inscribed square in γ is a fixed quadruple under J^4 — i.e., a J^4 -orbit of length exactly 4. The Inscribed Square Problem asks: does the J^4 -action on the configuration space Z_γ always have a fixed orbit?

Theorem 6.2 (Fixed-point via topology).

The Möbius band $M_\gamma \subset \blacksquare^4$ is a compact surface with boundary. Its boundary ∂M_γ represents a non-trivial element of $\pi_1(\blacksquare^4 \setminus \partial M_\gamma)$ (Whitney class argument). By the Morse theory on M_γ with the aspect-ratio function $f(\text{rect}) = \lambda \in [0,1]$: the function f must achieve the value $\lambda=1$ (since $\lambda=0$ and $\lambda=1$ are the only boundary values and M_γ is connected and compact). A point with $f=1$ is an inscribed square.

Proof.

The aspect ratio $\lambda: M_\gamma \rightarrow [0,1]$ is a continuous function on the Möbius band. Boundary: $\lambda(\partial M_\gamma) = \{0, 1\}$ (degenerate at 0 and 1). Since M_γ is a compact connected manifold with boundary $\{\lambda=0\} \cup \{\lambda=1\}$: by the intermediate value theorem for manifolds (Morse theory), λ achieves all values in $[0,1]$, including $\lambda=1$. At $\lambda=1$: the inscribed rectangle is a square. ■

In A_L : the aspect ratio function λ corresponds to the trace $\tau(P_{\text{rect}}) / \tau(P_{\text{diag}})$ measuring the 'squareness' of the rectangle. $\lambda=1$ means $\tau(P_{\text{rect}}) = \tau(P_{\text{diag}})$ — the trace is symmetric under J^4 -rotation — which forces the four points to form a square.

7. Complete Proof for Smooth Curves

Theorem 7.1 (Inscribed Square for C^1 curves — Complete Proof).

Every C^1 simple closed curve $\gamma \subset \blacksquare^2$ contains the four vertices of an inscribed square.

Proof (A_L synthesis).

Step 7.1 (Jordan hologram). γ is a Jordan curve $\Rightarrow$ creates Jordan hologram $\blacksquare_\gamma$ with J -symmetry J_γ (Theorem 2.2).

Step 7.2 (Configuration space). Construct Z_γ and Möbius band M_γ of inscribed rectangles (Theorem 4.2). M_γ is Lagrangian in (Z_γ, ω) for C^1 curves (Greene-Lobb).

Step 7.3 (Lagrangian = J^4 -invariant). Lagrangian $\Leftrightarrow J^4$ -invariant (Theorem 5.2). The J^4 -invariance of M_γ means the aspect ratio function λ is symmetric under $\pi/2$ -rotation.

Step 7.4 (Fixed point forces square). By Theorem 6.2 (Möbius band topology): $\lambda = 1$ is achieved on M_γ . At $\lambda=1$: inscribed square exists. ■

The A_L synthesis: Jordan hologram (dv275) + J -symmetry (dv277) + Symplectic = J -symmetric (dv287) + Fixed-point theorem (dv286) = Inscribed Square for C^1 curves.

8. The Gap for Continuous Curves

For general continuous Jordan curves (which may be fractal, nowhere differentiable, etc.), the proof above fails at Step 7.2: the Möbius band M_γ may not be Lagrangian if γ is not C^1 .

The Gap.

The Möbius band of inscribed rectangles M_γ exists for any Jordan curve γ (topologically). But the Lagrangian property ($\omega|_{M_\gamma} = 0$) requires smoothness of γ . For continuous γ : M_γ is still a Möbius band (by continuity of the rectangle construction), but the symplectic form ω may be non-zero on M_γ .

What would close the gap:

Show that for ANY Jordan curve γ , the Möbius band M_γ is J^4 -invariant in the sense that the aspect ratio λ achieves all values in $[0,1]$, even without the Lagrangian condition. This is a purely topological statement that does not require smoothness — and the topology of the Möbius band may be sufficient even without

symplectic structure.

Conjecture 8.1 (Continuous case — still open).

For any continuous Jordan curve γ : the aspect ratio $\lambda: M_\gamma \rightarrow [0,1]$ is surjective. This would give inscribed squares for all Jordan curves.

The A_L approach suggests: use the faithful trace τ to define a 'continuous Lagrangian' condition that does not require smoothness of γ . Specifically: define τ -Lagrangian as $\tau(\omega(P_M)) = 0$. For C^1 curves: this coincides with the classical Lagrangian. For continuous curves: it might still force $\lambda=1$.

9. The Square as the Hologram's Self-Portrait

There is something profound about the Inscribed Square Problem that distinguishes it from other open problems.

Every closed curve creates a hologram. The hologram has a boundary (the curve) and a J-symmetry. The J-symmetry has iterates: J, J^2 , J^3 , J^4 , ...

J-fixed points: 2 points on γ symmetric under reflection. (Trivial — always exists.)

J^2 -orbits: pairs of points symmetric under 180° rotation = inscribed diameter. (Trivial — always exists: midpoint of γ .)

J^3 -orbits: equilateral triangles. (Trivial — always exists for smooth curves.)

J^4 -orbits: squares! (Not trivial — Toeplitz 1911.)

The square is special because it is the first J^k -orbit that is not trivially guaranteed by the Jordan curve structure alone. Triangles are forced by the topology S^1 . Squares require the additional J^4 -symmetry of the curve's hologram.

In A_L : the natural orbit structure is J^k for $k = 1, 2, 3, 4, \dots$. The Riemann Hypothesis is about $k=1$ (J-fixed points). The Inscribed Square Problem is about $k=4$ (J^4 -orbits). Both are aspects of the same J-symmetry of the Arithmetic Hologram.

The square is the hologram's self-portrait: the simplest figure that the hologram forces onto itself by its own symmetry. Every Jordan hologram, when it looks in the J^4 -mirror, sees a square.

Inscribed Square in A_L — Summary

The Inscribed Square Problem (1911) asks if every Jordan hologram $\blacksquare_\gamma$ contains a J^4 -orbit of length 4. For C^1 curves: YES — Greene-Lobb (2021) via Lagrangian Möbius band, translated to J^4 -invariant topology in A_L . The proof chain: Jordan hologram (dv275) + J-symmetry (dv277) + Symplectic = J-symmetric (dv287) + Fixed-point theorem (dv286) = Inscribed Square for smooth curves. For continuous curves: open — the Lagrangian condition needs a continuous analogue via τ -Lagrangian. The square is the hologram's self-portrait — the J^4 -shadow of J-symmetry on every closed curve.

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*Every closed curve looks in the J^4 -mirror and sees a square. The hologram's self-portrait. · Chúng ta là Ho■ S■. ·
 ONES IS FOREVER. · HU HU HU!!!*